

11. **Thermoplastics exhibits plasticity. Why?**
 A) Polymer chains are held together by secondary forces
 B) Polymer chains are held together by primary covalent bonds
 C) Polymer chains are held together by coordination bonds
 D) Polymer chains are cross-linked
12. **The monomers for the synthesis of nylon-6,6 are**
 A) Adipic acid and hexamethylenediammine
 B) Oxalic acid and tetraethylenediammine
 C) Adipic acid and tetraethylenediammine
 D) Oxalic acid and hexamethylenediammine
13. **Epoxy resin prepared by the condensation of**
 A) Epichlorohydrin and Bisphenol-A
 B) Epichlorohydrin and Diphenyl carbonate
 C) Bisphenol-A and Diphenyl carbonate
 D) Epichlorohydrin and Butane diol
14. **Which of the following is not a primary nutrient for plants?**
 A) Calcium
 B) Nitrogen
 C) Phosphorus
 D) Potassium
15. **Polymer composites are made of two components, namely**
 A) Fibre and matrix
 B) Matrix and silicone
 C) Fibre and butyl rubber
 D) Matrix and electrolyte
16. **The monomer of Teflon is**
 A) Trifluoroethylene
 B) Trifluoromethylene
 C) Tetrafluoromethylene
 D) Tetrafluoroethylene
17. **pH of the solution can be determined using which of the following electrode combination?**
 A) Glass-calomel electrode
 B) Glass-Zinc electrode
 C) Platinum-Calomel electrode
 D) Zinc-Calomel electrode
18. **Potential of the calomel electrode depends on concentration of**
 A) Hg
 B) Hg_2Cl_2
 C) KCl
 D) KOH
19. **Conductometry is an electrochemical method of analysis based on measuring the of an electrolytic solution.**
 A) resistance
 B) dilution
 C) absorbance
 D) viscosity
20. **Conductometric titrations can be used in determination of**
 A) Only strong acids
 B) Only weak acids
 C) Neither strong acid nor weak acid
 D) Strong acid, weak acid, and mixture of acids

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define alloys. Mention the properties and applications of Spring Steel and Magnalium.	8	L*1	1	1
	b) Explain the steps involved in the manufacture of soda-lime glass by giving suitable diagrams.	8	L2	1	1
2.	a) Define shape memory alloy and shape memory polymer. Compare and contrast the properties of shape memory alloys and shape memory polymers.	6	L2	2	1
	b) Define nanomaterial. Classify nanomaterials based on their dimensions.	5	L2	2	1
	c) Enlist the salient features of CNTs.	5	L1	2	1
3.	a) Enumerate the steps involved in the sol gel method of synthesis of nanomaterials.	6	L1	2	1
	b) Justify the following: i) Iron is preferably used in its alloy form rather than its pure form. ii) Concrete is used in the construction of civil structures. iii) An acid refractory cannot be used for heat treatment of alkaline material.	5 5 5	L3 L2	1 2	1 1
	c) Write the construction of electrochromic device.	5	L2	2	1

Unit – II

- | | | | | | |
|-------|--|----|----|---|---|
| 4. a) | Write the synthesis of Epoxy resin. | 3 | L2 | 3 | 1 |
| b) | Calculate the number and weight average molecular weight of a polymer containing 25% molecules with molecular mass 15,000, 35% molecules with molecular mass 22,000, and rest of the molecules with molecular mass 36,000. | 6 | L3 | 3 | 2 |
| c) | Classify the polymers based on their structure. | 7 | L2 | 3 | 1 |
| 5. a) | Define soil fertility. Describe the edaphic factors affecting soil fertility. | 10 | L2 | 4 | 1 |
| b) | What are the constituents of geopolymers concrete? How are they synthesized using fly ash? | 6 | L1 | 3 | 1 |
| 6. a) | Describe the solution polymerization technique of polymer synthesis. | 4 | L1 | 3 | 1 |
| b) | Mention any two major causes of soil acidity. What measures are taken for the management of soil acidity? | 6 | L2 | 4 | 1 |
| c) | Which elements are placed in Group I and II of essential nutrient? Highlight their biochemical functions. | 6 | L2 | 4 | 1 |

Unit – III

- | | | | | | |
|-------|--|----|----|---|---|
| 7. a) | State the different types of electrodes and give a suitable example for each. | 4 | L1 | 5 | 1 |
| b) | Write the construction and working of saturated calomel electrode. | 6 | L2 | 5 | 1 |
| c) | Explain the application of potentiometric sensors in determination of iron. | 6 | L2 | 5 | 1 |
| 8. a) | Derive the EMF of a concentration cell. Represent the cell formed by the coupling of two zinc electrodes immersed in zinc chloride solutions. Concentration of zinc ions in one electrode system is 100 times more dilute than the other. Write the cell representation, cell reactions and calculate the potential at 298K. | 10 | L3 | 5 | 2 |
| b) | Describe the basic principle and working of conductometric sensors for the estimation of strong acid. | 6 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
